Salary Prediction

Employee Salary Prediction Using Machine Learning

Title:
Employee Salary Prediction using Machine Learning

Problem Statement:

Organizations need to estimate employee salaries based on various factors like experience, education, job role, and location. Manual estimation may lead to bias or inconsistency. This project aims to build a machine learning model that accurately predicts employee salaries using historical data.

Explanation of the Project

This project focuses on predicting salaries using supervised machine learning techniques.

- Input features: Experience, education level, job role, location, etc.
- Output: Predicted salary
- The system learns patterns from past employee data and predicts salary for new inputs.

Steps involved:

1. Data collection
2. Data preprocessing
3. Feature selection
4. Model training
5. Evaluation
6. Prediction

3. Description of the Model/Algorithm(s) Used

You can mention 2–3 algorithm

1. Linear Regression

Used for predicting continuous values

1. **Finds relationship between independent variables and salary**
2. **Simple and easy to interpret**

- **2. Decision Tree**

3. **Splits data into branches based on conditions**
4. **Works well with non-linear data**
5. **Easy to visualize**

- **3. Random Forest (Optional – Best Model)**

6. **Combination of multiple decision trees**
7. **Reduces overfitting**
8. **Gives higher accuracy**

4. Accuracy / Performance Metrics of the Model

Common metrics used:

- Mean Absolute Error (MAE)
- → Average difference between predicted and actual salary
- Mean Squared Error (MSE)
- → Penalizes large errors
- R^2 Score (Accuracy)
- → Shows how well the model fits data
- (Closer to 1 = better performance)
- Example:
- MAE = 2500
- MSE = 1,200,000
- R^2 Score = 0.85 (85% accuracy)

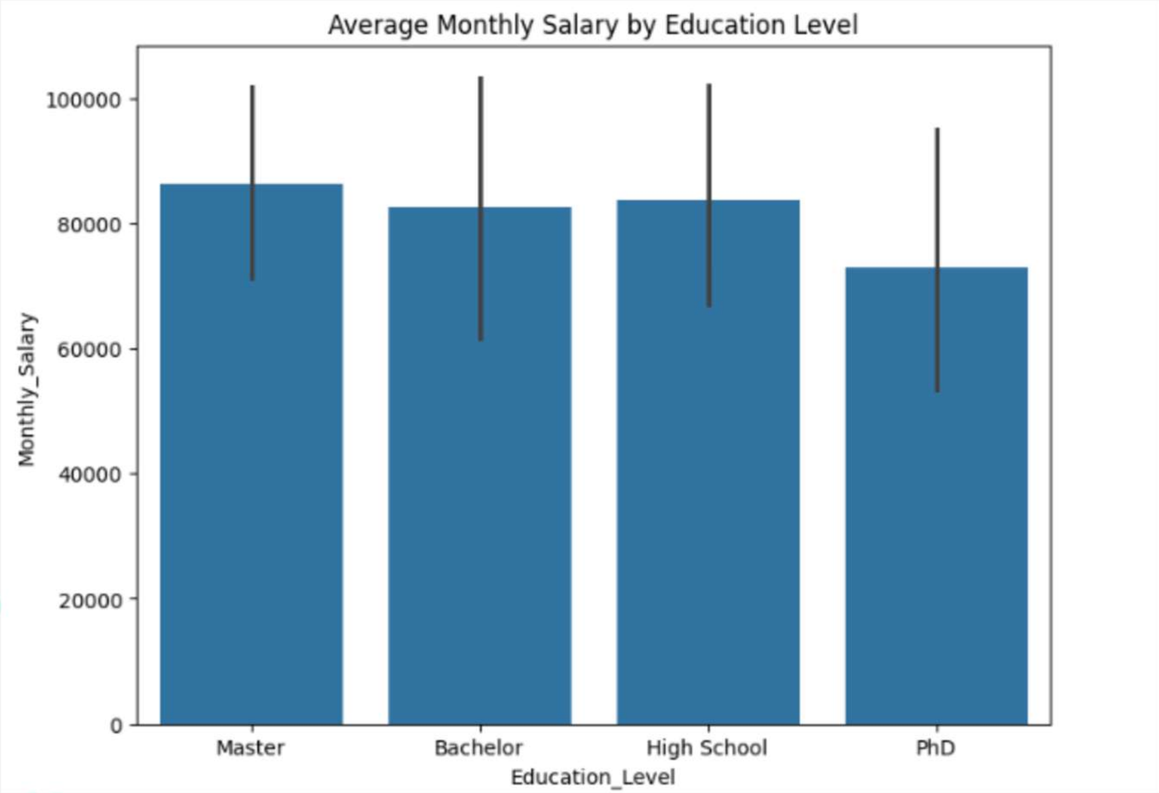
5. Figures/Plots Showing Results

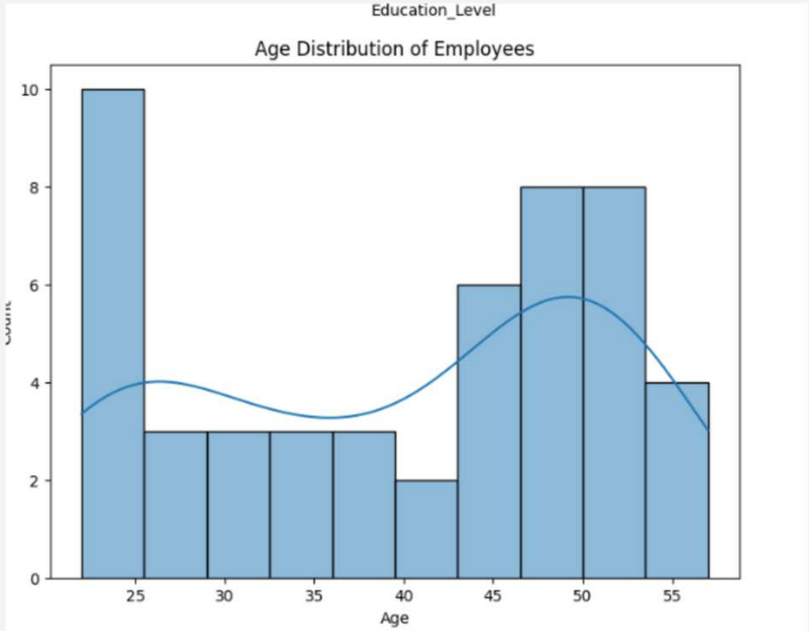
- **Include these graphs in your PPT:**

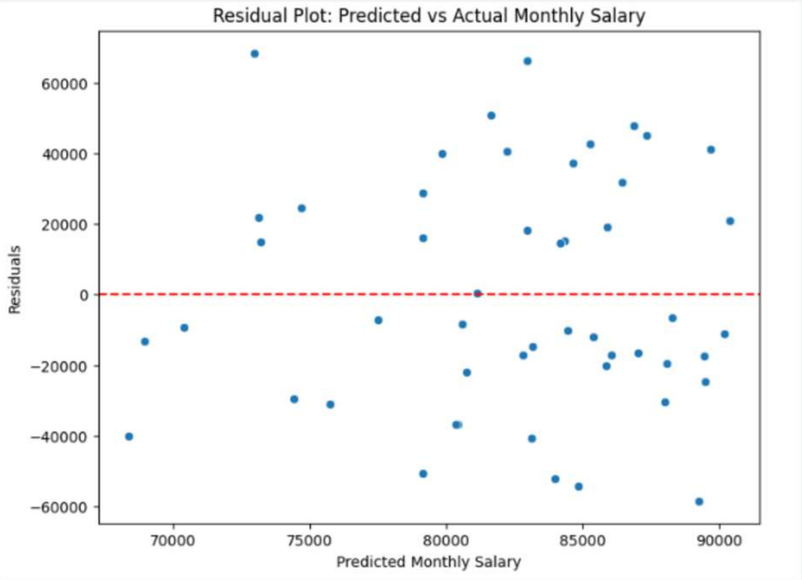
1. **Scatter Plot (Actual vs Predicted Salary)**
2. **Regression Line Graph**

- **Feature Importance Graph (for Random Forest)**

3. **Histogram of Salary Distribution**


Average Monthly Salary by Education Level
Monthly_Salary
100000
80000
60000
40000
20000
0
Master
Bachelor
High School
PhD
Education_Level


Education_Level
Age Distribution of Employees
Count
Age
0
2
4
6
8
10
25
30
35
40
45
50
55

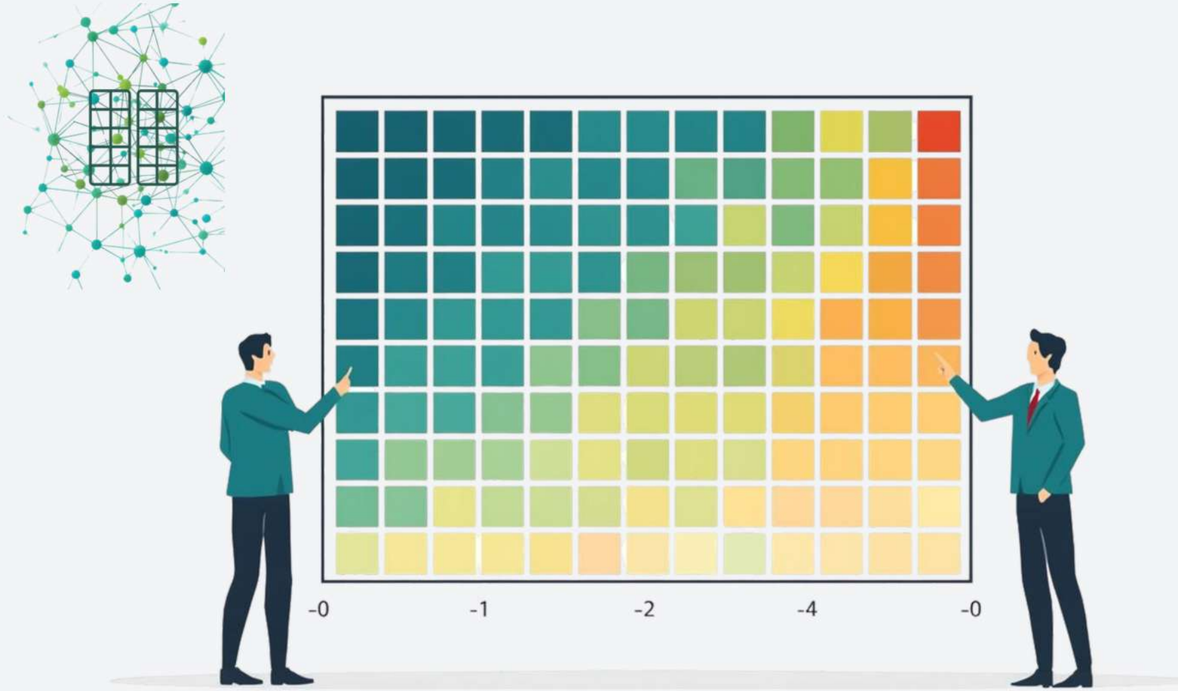

Residual Plot: Predicted vs Actual Monthly Salary
Residuals
Predicted Monthly Salary
60000
40000
20000
0
−20000
−40000
−60000
70000
75000
80000
85000
90000

- **6. Outcome & Conclusion of the Project**

- **Outcome:**
- **Successfully built a model that predicts employee salaries**
- **Helps HR departments make data-driven decisions**
- **Conclusion:**
- **Machine learning improves accuracy and reduces bias**
- **Random Forest performed better compared to other models**
- **Model can be improved using more data and features**

7. GitHub Repository Link

https://github.com/2403a51178/1178/blob/main/ML_Project.ipynb



Thank You!